2.4 Sum and Difference Formulas

Name _____ Period ____ Date _____

Pre-Calculus 2.4 (30-minute plan) · Student sheet

I can…

- use sum and difference formulas to find exact values.
- use sum and difference formulas to rewrite and evaluate expressions.
- use sum and difference formulas to solve equations.

Part 1 · Notes: fill in the blanks

$\sin(u + v)$ = _____ $\sin(u - v)$ = _____

$\cos(u + v)$ = _____ $\cos(u - v)$ = _____

$\tan(u + v)$ = _____ $\tan(u - v)$ = _____

In the sine formulas the sign _____; in the cosine formulas the sign _____.

Part 2 · Examples (work them with the class)

Example 1

Find the exact value of $\sin 75°$.

Example 2

Write $\sin 42° \cos 12° - \cos 42° \sin 12°$ as a trigonometric function of one angle. Then find its exact value.

Example 3

$\sin u = \frac{5}{13}$ with $0 < u < \frac{\pi}{2}$, and $\cos v = -\frac{3}{5}$ with $\frac{\pi}{2} < v < \pi$. Find the exact value of $\sin(u + v)$.

Part 3 · Partner practice

Partner practice

Find the exact value of cos 105°.